

Disordered $J - J'$ Heisenberg Antiferromagnet on a Triangular Lattice

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The effect of quenched bond disorder in the triangular lattice Heisenberg antiferromagnet has attracted a lot of attention lately. Recent experiments on crystals which are believed to realize this model typically suggest emergence of a glassy phase when the exchange interactions between the spin degrees of freedom are sufficiently disordered. It is also understood that in the presence of bond disorder the 120° non-collinear chiral order is destroyed in $d = 2$ lower critical dimension, although, with a correlation length that scales exponentially with the inverse power of the width of the bond disorder. This rules out the possibility to characterize the system without carefully eliminating the finite size effects. To this end we resort to two different microscopic modelling of the disordered $J_1 - J_2$ Heisenberg hamiltonian, improving on the system size limitations of prior calculations and also mapping out the entire phase diagram of the model which corroborates to the signatures observed in experiments.

I. INTRODUCTION

The heisenberg antiferromagnet in a triangular lattice has long been a model of interest as it realizes the simplest case of geometric frustration in a spin model. The quantum mechanical next nearest neighbour $J_1 - J_2$ Heisenberg hamiltonian in thought to produce an interesting phase diagram with a Z_2 spin liquid phase interspersed between the non-collinear 120° spiral and a stripy magnetic order.

II. THE DISORDERED $J - J'$ HEISENBERG HAMILTONIAN ON A TRIANGULAR LATTICE

The disordered $J - J'$ antiferromagnetic Heisenberg model in the presence of an external field can be simply expressed in terms of the bilinears of spin operators,

$$H = \sum_{\langle i,j \rangle} J_{i,j} \vec{S}_i \cdot \vec{S}_j + \sum_{\langle\langle i,j \rangle\rangle} \alpha J'_{i,j} \vec{S}_i \cdot \vec{S}_j + \sum_i \vec{h}_i \cdot \vec{S}_i \quad (1)$$

where the symbols $\langle i,j \rangle$ and $\langle\langle i,j \rangle\rangle$ represents sum over nearest and next to nearest neighbours in the triangular lattice. In a clean system all the couplings will be fixed to a scale and in the absence of the external field $\vec{h}$, α becomes the sole tunable parameter that determines the relative strength of the next to nearest neighbour (NNN) couplings over the nearest neighbour (NN) couplings. However, in a disordered lattice the couplings will be random and this can be simulated by having

J and J' drawn from a random distribution $P(J, \Delta)$ with Δ characterizing the width of the disorder distribution. Δ is also called the disorder strength as the wider the distribution gets the less likely it becomes to find a neighbourhood of uniform couplings in the lattice. The parameter α now underscores the relative strength between the average NNN couplings and the average NN couplings.

The clean zero field limit ($\Delta = 0, \vec{h} = 0$) has been studied extensively using various microscopic and field theoretic methods. Classically, it can be shown that the ground state of the hamiltonian in this limit goes through a phase transition from the 120° non-collinear magnetic order with an ordering wave vector $\vec{Q} = (2\pi/3, 0)$ to a collinear stripe order with $\vec{Q} = (x, x)$.

III. SPINWAVE EXCITATIONS WITH FINITE DISORDER

A. Zero magnetic field

B. Large magnetic field

IV. MEAN-FIELD THEORY IN REAL SPACE

V. RESULTS

VI. CONCLUSIONS